

Calculated Metrics:

- Completion Time (CT): The absolute time on the timeline when a process completely finishes its execution.
- Waiting Time (WT): The total time a process spends waiting in the ready queue before and during execution. Calculated as $WT = CT - \text{Arrival Time} - \text{Burst Time}$.
- Turnaround Time (TAT): The total time taken from the process's arrival to its complete execution. Calculated as $TAT = CT - \text{Arrival Time}$.
- Response Time (RT): The time elapsed from when a process arrives to the first time it is allocated the CPU. Calculated as $RT = \text{Start Time} - \text{Arrival Time}$.
- Averages: The system computes the Average Waiting Time, Average Turnaround Time, and Average Response Time across all processes to determine the overall efficiency of the algorithm.